Ex no 11

10/10/24

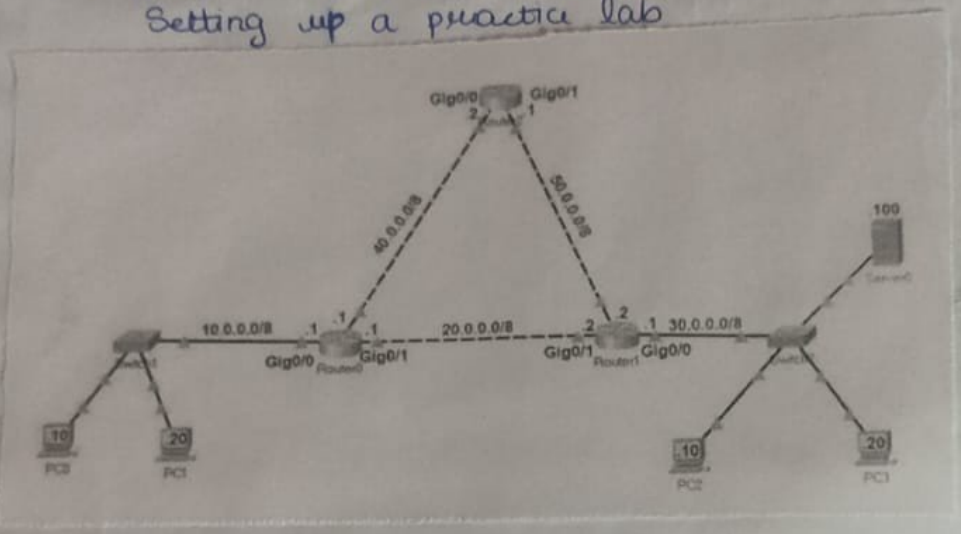
Experiment - 11 (A)

Aim:

Simulate static routing configuration using cisco packet tracer

Procedure

Setting up a practice lab



Creating, adding, verifying static router connected network of each router

Router	Available on local	Available on other router
Router 0	10.0.0.0/8 20.0.0.0/8	30.0.0.0/8, 50.0.0.0/8
Router 1	20.0.0.0/8, 30.0.0.0/8	10.0.0.0/8 40.0.0.0/8

Router 0 Configuration :

Router 0 enable

Router # configure terminal

Router(config) # ip route 30.0.0.0 255.0.0.0
20.0.0.2

Router(config) # ip route 30.0.0.0 255.0.0.0
40.0.0.2

Router(config) # exit

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2 10 Primary route
Router(config)#ip route 30.0.0.0 255.0.0.0 40.0.0.2 20 Backup route
Router(config)#ip route 30.0.0.100 255.255.255.255 40.0.0.2 10 Primary route
Router(config)#ip route 30.0.0.100 255.255.255.255 20.0.0.2 20 Backup route
Router(config)#ip route 50.0.0.0 255.0.0.0 40.0.0.2 10 Primary route
Router(config)#ip route 50.0.0.0 255.0.0.0 20.0.0.2 20 Backup route
Router(config)#exit
Router#show ip route static
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S       30.0.0.0/8 [10/0] via 20.0.0.2
S       30.0.0.100/32 [10/0] via 40.0.0.2
S    50.0.0.0/8 [10/0] via 40.0.0.2

Router#
```

Router adds only primary routes to the routing table.

Router 1 Configuration :

Router > enable

Router # configure terminal

Router (config) # ip route 10.0.0.0 255.0.0.0 20.0.0.1D

Router (config) # ip route 10.0.0.0 255.0.0.0 50.0.0.20

Router (config) # exit

Router2

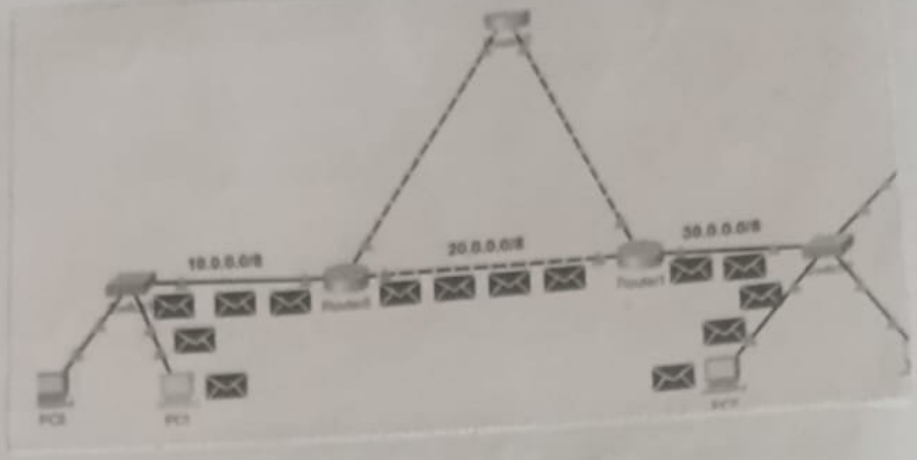
Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with
Router(config)#ip route 10.0.0.0 255.0.0.0 40.0.0.1
Router(config)#ip route 30.0.0.0 255.0.0.0 50.0.0.2
Router(config)#exit
Router#show ip route static
S    10.0.0.0/8 [1/0] via 40.0.0.1
S    30.0.0.0/8 [1/0] via 50.0.0.2

Router#
```

Verifying static routing



To simulate the failure of main route you can delete the link btwn router 0 and router 1

```
Command Prompt

Packet Tracer PC Command Line 1.0
PC>ipconfig

FastEthernet0 Connection:(default port)
Link-local IPv6 Address.....: FE80::260:70F
IP Address.....: 20.0.0.2
Subnet Mask.....: 255.0.0.0
Default Gateway.....: 20.0.0.1

PC>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 10.0.0.2: bytes=32 time=3ms TTL=126
Reply from 10.0.0.2: bytes=32 time=3ms TTL=126
Reply from 10.0.0.2: bytes=32 time=3ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 3ms, Average = 3ms

PC>
```

Result:

The experiment has been completed successfully

10/10/24

Experiment-11(B)

Aim:

To simulate RIP using Cisco Packet Tracer

Initial IP configuration

Device	Interface	IP config	connected with
PC0	FastEthernet	10.0.0.2/8	Router0's Fa 0/1
Router0	Fa 0/9	10.0.0.1/8	PC0's fast Ethernet
Router0	S0/0/1	192.168.1.254/30	Router 2's S0/0/1
Router1	S0/0/1	192.168.1.253/30	Router 0's S0/0/0
Router1	S0/0/1	192.168.1.256/30	Router 0's S0/0/1
PC1	Fast ethernet	20002/30	Router 2's Fa 0/1

Assign IP addresses to PC's

Double click PC's and click desktop menu item and click IP config

Router 0:

Router (config) # interface fast Ethernet 0/0

Router (config-if) # ip address 10.0.0.1 255.0.0.0 Router (config) # no Shutdown

Router (config-if) # exit

Router 1:

Router, enable

Router # configure

Router (config) # interface serial 0/0/0

Router (config-if) # no shutdown

Router (config-if) # exit

Router 2:

Router> enable
Router # configure
terminal
Router (config) # interface
Router (config-if) # ip address 200.0.11
255.0.0.0 Router (config-if) #no shutdown
Router (config-if) #exit

Access the command prompt of PC1 and use ping command to test connectivity from PC0

Result:

The experiment has been computed succes